

## Automobile exhaust as a source of $^{13}\text{C}$ - and D-enriched atmospheric methane in urban areas

Fumiko Nakagawa <sup>a,\*</sup>, Urumu Tsunogai <sup>a</sup>, Daisuke D. Komatsu <sup>a</sup>,  
Keita Yamada <sup>b,d</sup>, Naohiro Yoshida <sup>b,c,d</sup>, Jun Moriizumi <sup>e</sup>,  
Koichiro Nagamine <sup>e,1</sup>, Takao Iida <sup>e</sup>, Yukimasa Ikebe <sup>e</sup>

<sup>a</sup> Division of Earth and Planetary Sciences, Graduate School of Science, Hokkaido University, N10 W8 Kita-Ku, Sapporo, Hokkaido 060-0810, Japan

<sup>b</sup> Department of Environmental Chemistry and Engineering, Interdisciplinary Graduate School of Science and Engineering, Tokyo Institute of Technology, 4259 Nagatsuta, Midori-ku, Yokohama, Kanagawa 226-8502, Japan

<sup>c</sup> Department of Environmental Science and Technology, Interdisciplinary Graduate School of Science and Engineering, Tokyo Institute of Technology, 4259 Nagatsuta, Midori-ku, Yokohama 226-8502, Japan

<sup>d</sup> SORST Project, Japan Science and Technology Corporation, Kawaguchi, Saitama 332-0012, Japan

<sup>e</sup> Department of Nuclear Engineering, Graduate School of Engineering, Nagoya University, Furo-cho, Chikusa-ku, Nagoya 464-8603, Japan

Received 21 March 2004; accepted 13 January 2005

(returned to author for revision 20 September 2004)

Available online 2 April 2005

### Abstract

The stable carbon and hydrogen isotopic compositions ( $\delta^{13}\text{C}$  and  $\delta\text{D}$ , respectively) of methane ( $\text{CH}_4$ ) in automobile exhaust were determined in order to quantify by mass balance the contribution from this source to atmospheric  $\text{CH}_4$  in an urban area of Nagoya, Japan. Both  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of  $\text{CH}_4$  in automobile exhaust increased in accordance with the model year of the engine, probably as a result of isotope fractionation effects associated with oxidation over metal catalysts in the catalytic converter of modern vehicles. Thus,  $\text{CH}_4$  emissions from recent automobiles exhibit conspicuous  $^{13}\text{C}$  and D enrichment compared to the flux from other major anthropogenic sources, such as natural gas leakage, landfills and rice paddies. Using average  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values estimated for local sources, automobile exhaust in Nagoya, Japan, was determined to contribute significant amounts (up to 30%) of  $\text{CH}_4$  to the troposphere in the study area.

© 2005 Elsevier Ltd. All rights reserved.

\* Corresponding author. Tel.: +81 11 706 2731; fax: +81 11 746 0394.

E-mail address: [fuminaka@ep.sci.hokudai.ac.jp](mailto:fuminaka@ep.sci.hokudai.ac.jp) (F. Nakagawa).

<sup>1</sup> Present address: Department of Complex Systems Science, Graduate School of Information Science, Nagoya University, Furo-cho, Chikusa-ku, Nagoya 464-8601, Japan.

### 1. Introduction

#### 1.1. Background

Methane ( $\text{CH}_4$ ) is an important trace gas in the Earth's atmosphere, both as a greenhouse gas and as a key molecule in photochemical reactions in the

troposphere and stratosphere (Cicerone and Oremland, 1988). During the past 300 years, the abundance of atmospheric CH<sub>4</sub> has increased rapidly, more than doubling to its present concentration level of 1.73 ppmv (Craig and Chou, 1982; Etheridge et al., 1998; Pearman et al., 1986; Rasmussen and Khalil, 1984; Stauffer et al., 1985), possibly contributing significantly to changes in atmospheric chemistry and to global warming. The recent enhancement in atmospheric CH<sub>4</sub> levels has resulted mainly from emissions from anthropogenic sources, including natural gas wells, coal mines, the petroleum industry, enteric fermentation in ruminants, rice paddies, landfills, animal wastes, domestic sewage, biomass burning and a few other minor sources (IPCC, 1995; Lacroix, 1993). In order to develop suitable strategies for controlling CH<sub>4</sub> emissions to the atmosphere, the strength of anthropogenic sources needs to be accurately assessed.

Highly populated urban areas host many anthropogenic sources of CH<sub>4</sub> and, as a result, accumulation of CH<sub>4</sub> emitted from local sources (within several tens of kilometers) typically occurs at night during nocturnal stratification of the atmosphere. During the daytime, the local concentration of atmosphere CH<sub>4</sub> usually returns to the background level normally found in the free troposphere (Lowry et al., 2001; Moriizumi et al., 1996). The magnitude of anthropogenic CH<sub>4</sub> emissions in urban environments shows rapid and large temporal variations and trends on a scale of a few years to decades (Ito et al., 2000; Lowry et al., 2001) because of changes in parameters, such as population size and distribution, types of energy resources, methods of waste disposal, and lifestyle. It is important to clarify the main temporal contributor to urban atmospheric CH<sub>4</sub> because it may help to predict future emission levels in a given urban area.

Automobile exhaust is one of the known sources of atmospheric CH<sub>4</sub> in urban environments, but land use in such areas generally is too complex to allow local contributions from automobiles to be estimated by measurement of the CH<sub>4</sub> tropospheric mixing ratio at a limited number of fixed stations. However, the stable isotopic composition of CH<sub>4</sub> is a useful tracer for identifying emission sources and constraining flux strength at both the local (Kuc et al., 2003; Levin et al., 1993; Yamada et al., 2003) and global scale (Quay et al., 1999) via isotope mass balance models. However, evaluating source strengths of atmospheric CH<sub>4</sub> using <sup>13</sup>C and D as tracers requires a knowledge of  $\delta^{13}\text{C}$  and  $\delta\text{D}$  CH<sub>4</sub> values from representative emission sources. To date,  $\delta\text{D}$  values of CH<sub>4</sub> from automobile exhaust have yet to be reported and to our knowledge, pioneering studies have been conducted only for <sup>13</sup>C (Chanton et al., 2000; Levin et al., 1999). Consequently, we determined  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of CH<sub>4</sub> from automobile exhaust in Nagoya City to evaluate the contribution of

automobile emissions to tropospheric CH<sub>4</sub> levels in an urban area using isotope mass balance. Other local sources include natural gas leakage, rice paddies and a landfill.

## 1.2. Study area

The evaluation of automobile exhaust in an urban atmosphere was conducted in Nagoya City, the fourth largest city in Japan, having a population of more than two million. In the city, temporal variations in atmospheric CH<sub>4</sub> abundance have been monitored at several stations (Ito et al., 2000, 2001; Moriizumi et al., 1996). These studies indicated that urban areas in Nagoya are a significant net source of atmospheric CH<sub>4</sub>, having an areal flux ranging from 0.01 to 0.02 g CH<sub>4</sub> m<sup>-2</sup> d<sup>-1</sup> (Moriizumi et al., 1996). By comparing spatial and temporal variations in urban atmospheric CH<sub>4</sub> with meteorological conditions, it has been demonstrated that local landfills contribute significant amounts of CH<sub>4</sub> to the urban atmosphere in Nagoya (Ito et al., 2001). In another study, which measured the <sup>14</sup>C content of tropospheric CH<sub>4</sub> to estimate the contribution of fossil (<sup>14</sup>C-free) sources to the urban atmosphere, it was concluded that natural gas was the dominant source of atmospheric CH<sub>4</sub> in urban areas (Moriizumi et al., 1998). However, the contribution of automobile exhaust, which is another possible source of fossil CH<sub>4</sub> emissions to the atmosphere, has yet to be evaluated and is the focus of this study. In addition, a suburban area of Nagoya City was studied for comparison.

## 2. Methods

### 2.1. Atmospheric samples

Air sampling was performed at three sites in Nagoya City: Higashi-Sakura (Stn-H), Tomida (Stn-T), and Nagoya University (Stn-N; Fig. 1). Stn-H is located in downtown Nagoya City. Stn-T is situated in a suburban area where rice paddies are farmed near houses. Stn-N is located on the eastern side of the city, which consists mainly of residences.

Air samples were collected at Stn-H in the winter (November and December) of 1993, at Stn-T in summer (July and August) of 1994 and at Stn-N in both summer (July 1993) and winter (November 1993 and January and February 1994). Samples were collected both before sunrise and during the daytime in order to quantify variations in the isotopic composition of atmospheric CH<sub>4</sub> associated with its diurnal fluctuation in abundance. At Stn-H and Stn-T, air from a height of ~1.5 m above the ground was pumped into a large Tedlar bag (polyvinyl fluoride; 1.1 m<sup>3</sup> STP). At Stn-N, a similar volume of air was sampled at a height of ~10 m using the same

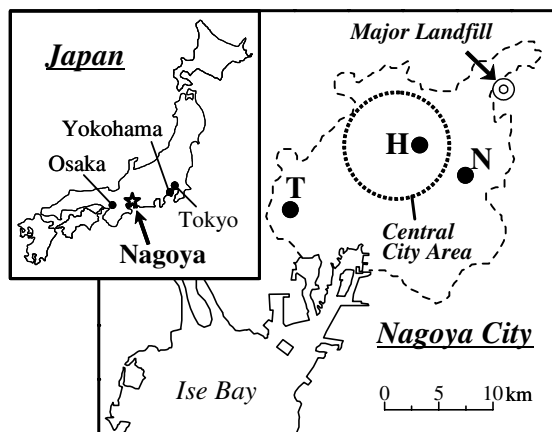


Fig. 1. Location of air sample collection sites in Nagoya, Japan: Higashi-Sakura (H), representing an urban area and Tomida (T) and Nagoya University (N), representing suburban areas.

procedure. Collection of each sample took approximately 20 min.

Methane in air samples was extracted using the method of Moriizumi et al. (1995). The samples were drawn at a flow rate of 2.5 or 3.5 L/min into a vacuum line containing several cryogenic traps, including two liquid nitrogen traps, an active charcoal trap, and a 5A molecular sieve (MS-5A) trap. Samples were processed at a pressure lower than  $2 \times 10^4$  Pa to prevent liquefaction of oxygen. After removal of  $\text{H}_2\text{O}$  and  $\text{CO}_2$  in the liquid nitrogen traps, the samples were passed through a trap filled with granular activated carbon cooled with liquid nitrogen to absorb  $\text{CH}_4$ . Atmospheric nitrogen ( $\text{N}_2$ ) and oxygen ( $\text{O}_2$ ) were also absorbed on the activated carbon; however, these gases were removed subsequently by evacuation of the trap at dry ice/ethanol temperature. The remaining sample gas trapped on the activated carbon, which consisted of  $\text{CH}_4$  and  $\sim 20$  ml of air, was desorbed by heating the trap to  $260^\circ\text{C}$ . The sample was then adsorbed at liquid nitrogen temperature on 0.5 g of MS-5A pellets contained in a tube. The adsorbed sample gas was desorbed in the tube at room temperature and introduced via helium carrier gas (99.9999% purity,  $1.5 \times 10^5$  Pa) into a gas chromatograph to separate and purify the  $\text{CH}_4$ . The gas chromatograph (Ohkura model-802T, Tokyo, Japan) was equipped with a thermal conductivity detector (TCD) and an MS-5A (60/80 mesh) column. The purified  $\text{CH}_4$  was carried directly into a CuO combustion tube ( $830^\circ\text{C}$ ) to oxidize it to  $\text{CO}_2$  and  $\text{H}_2\text{O}$ . The  $\text{H}_2\text{O}$  was converted to  $\text{H}_2$  for stable hydrogen isotope analysis using the zinc reduction method of Coleman et al. (1982). The overall yield of  $\text{CH}_4$  from air samples typically was greater than 95%.

Stable carbon and hydrogen isotope analyses were performed using a Finnigan MAT 252 isotope ratio mass spectrometer (IRMS). Values are expressed in per mil (‰) according to the equation:

$$\delta(\text{‰}) = ((R_{\text{sample}}/R_{\text{standard}}) - 1) \times 1000, \quad (1)$$

where R is either  $^{13}\text{C}/^{12}\text{C}$  or D/H. Values are reported relative to V PDB for  $\delta^{13}\text{C}$  and V SMOW for  $\delta\text{D}$ . The overall precision of the  $\delta^{13}\text{C}$  and  $\delta\text{D}$  analyses was  $\pm 0.1\text{‰}$  and  $\pm 1.5\text{‰}$ , respectively.

## 2.2. Automobile exhaust

Automobile exhaust was collected from the tailpipes of idling automobiles to determine  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of  $\text{CH}_4$  in exhaust gases. Three gasoline automobiles of different age were selected to evaluate the influence of engine combustion efficiency and/or the effects of catalytic converters in the sample vehicles: (1) 1966 Volkswagen Type 1 (CAR-66), a very old automobile without a catalytic converter; (2) 1993 Nissan Primera (CAR-93), a recent automobile with a catalytic converter; (3) 2002 Honda Fit (CAR-02), a brand new automobile with a catalytic converter. Automobile exhaust was sampled into glass bottles (200 ml) fitted with inlet and outlet stopcocks. The inlet was connected to the tail pipe using Tygon® tube (6-mm i.d.  $\times$   $\sim 0.3$ -m length). After purging the glass bottle with exhaust gases, the two stopcocks were closed. Seven to nine gas samples were taken from each automobile.

The concentration and stable carbon isotope composition of  $\text{CH}_4$  were measured using a combination of two previously reported methods: (1) a system for gas sample introduction comprised a vacuum line equipped with a 6-port valve and a pre-concentration column (Tsunogai et al., 2002) containing silica gel (instead of silica gel and MS-5A in the cold trap) and (2) a more sensitive analytical system for  $\text{CH}_4$  analysis (Tsunogai et al., 2000) using gas chromatography-combustion-IRMS. In system 1, a gas sample of appropriate volume was introduced into the vacuum line and collected on the pre-concentration column which was cooled in liquid nitrogen. The pre-concentration column was then heated to approximately  $90^\circ\text{C}$  using hot water while being flushed with ultra pure helium. After removal of  $\text{H}_2\text{O}$  and  $\text{CO}_2$  using a magnesium perchlorate/Ascarite® trap, the remaining gas components were passed through a liquid oxygen temperature ( $-183^\circ\text{C}$ ) mini GC (10-mm long column packed with Porapak Q) to separate  $\text{CH}_4$  from other sample gases. The purified  $\text{CH}_4$  was injected into a Poraplot-Q capillary column at liquid oxygen temperature to concentrate the  $\text{CH}_4$  at the head of the column. The column head was then rapidly heated to  $80^\circ\text{C}$  under a continuous flow of helium at 3.0 ml/min. Column-separated  $\text{CH}_4$  was oxidized to  $\text{CO}_2$  at  $960^\circ\text{C}$  in a combustion furnace containing CuO/Pt catalyst

and introduced into a Finnigan MAT 252 isotope ratio mass spectrometer for concentration and  $^{13}\text{C}/^{12}\text{C}$  analysis. The relative error in the determination of  $\text{CH}_4$  concentration by this method was less than 3% and the overall precision of  $\delta^{13}\text{C}$  analysis was  $\pm 0.2\text{‰}$ .

The stable hydrogen isotope composition of  $\text{CH}_4$  was determined using a gas chromatography/high-temperature conversion/isotope ratio mass spectrometry (GC/TC/IRMS) system (Yamada et al., 2003). A gas sample approximately 10 times the size required for  $\delta^{13}\text{C}$  analysis was introduced into a vacuum line and collected on a pre-concentration column (active carbon 30/60 mesh) at liquid nitrogen temperature. The pre-concentration column was heated to dry ice/ethanol temperature while atmospheric  $\text{N}_2$  and  $\text{O}_2$ , which were also absorbed on the activated carbon, were removed by pumping. The  $\text{CH}_4$  and other remaining gas components were transferred to a second concentration trap (10-mm long column packed with Porapak Q at liquid nitrogen temperature) by slowly heating the pre-concentration column to room temperature while flushing it with ultra pure helium. During the transfer procedure,  $\text{H}_2\text{O}$  and  $\text{CO}_2$  were removed by passage through a magnesium

perchlorate/Ascarite® trap. Methane was separated completely from  $\text{CO}$  and mostly from  $\text{N}_2$  and  $\text{O}_2$  by passing the sample through a MS-5A column (30/60 mesh). The resulting  $\text{CH}_4$  plus small amounts of remaining gases were introduced into a Hewlett–Packard PLOT-Q capillary column by heating the Porapak-Q trap to room temperature. Column-separated  $\text{CH}_4$  was pyrolyzed to  $\text{H}_2$  in a ceramic tube ( $\text{Al}_2\text{O}_3$ ) furnace at  $1440^\circ\text{C}$  and introduced into a Delta<sup>plus</sup>XL, (Thermo Electron) IRMS for D/H analysis. The overall precision of  $\delta\text{D}$  analysis using this method was  $\pm 3\text{‰}$ .

### 3. Results and discussion

#### 3.1. Stable isotope compositions of $\text{CH}_4$ from automobile exhaust and other local sources

##### 3.1.1. Automobile exhaust

The concentration and  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of  $\text{CH}_4$  for 25 exhaust gas samples collected from the three automobiles are listed in Table 1. Exhaust gas from the newer automobiles had a lower  $\text{CH}_4$  content

Table 1

Concentration and stable carbon and hydrogen isotope compositions of  $\text{CH}_4$  in automobile exhaust

| Sample                                 |      |      | $\text{CH}_4$         |                  |
|----------------------------------------|------|------|-----------------------|------------------|
| No.                                    | rpm  | ppmv | $\delta^{13}\text{C}$ | $\delta\text{D}$ |
| <i>CAR-66 (1966 Volkswagen TYPE-1)</i> |      |      |                       |                  |
| V-1                                    | 1000 | 175  | −21.5                 | −163             |
| V-2                                    | 1000 | 1068 | −13.4                 | −125             |
| V-3                                    | 1000 | 917  | −12.9                 | −139             |
| V-4                                    | 2000 | 223  | −21.7                 | −169             |
| V-5                                    | 2000 | 216  | −20.9                 | −163             |
| V-6                                    | 2000 | 235  | −21.5                 | −170             |
| V-7                                    | 3000 | 267  | −22.3                 | −174             |
| V-8                                    | 3000 | 265  | −22.8                 | −177             |
| V-9                                    | 3000 | 219  | −22.8                 | −178             |
| <i>CAR-93 (1993 Nissan PRIMERA)</i>    |      |      |                       |                  |
| N-1                                    | 800  | 120  | −13.6                 | −129             |
| N-2                                    | 1300 | 61   | −9.6                  | −119             |
| N-3                                    | 1500 | 42   | −9.4                  | −130             |
| N-4                                    | 1800 | 46   | −11.9                 | −120             |
| N-5                                    | 1800 | 120  | −13.6                 | −124             |
| N-6                                    | 2000 | 46   | −8.6                  | −106             |
| N-7                                    | 2600 | 45   | −11.9                 | −134             |
| <i>CAR-02 (2002 Honda FIT)</i>         |      |      |                       |                  |
| H-1                                    | 1000 | 46   | −12.7                 | −121             |
| H-2                                    | 1000 | 18   | −3.9                  | −38              |
| H-3                                    | 1000 | 32   | −5.9                  | −78              |
| H-4                                    | 2000 | 49   | −5.5                  | −21              |
| H-5                                    | 2000 | 76   | −11.6                 | −93              |
| H-6                                    | 2000 | 78   | −11.6                 | −101             |
| H-7                                    | 3000 | 79   | −10.9                 | −94              |
| H-8                                    | 3000 | 84   | −10.9                 | −94              |
| H-9                                    | 3000 | 89   | −11.3                 | −93              |

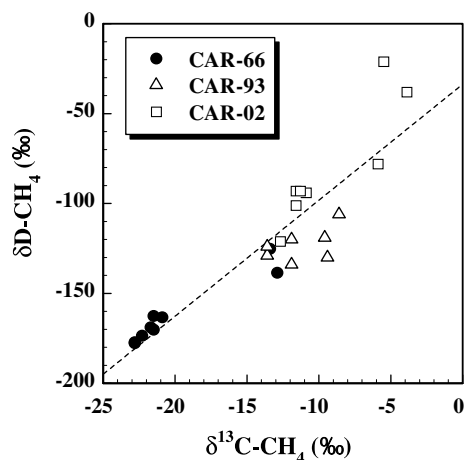


Fig. 2. Relationship between stable carbon and hydrogen isotope compositions of  $\text{CH}_4$  in automobile exhaust.

( $69 \pm 36$  ppmv for CAR-93 and  $61 \pm 25$  ppmv for CAR-02) compared to that from the older automobile ( $229 \pm 32$  ppmv for CAR-66; two abnormally high values were excluded from this average). The difference probably results from more effective combustion within the newer engines and/or the lack of a catalytic converter in CAR-66.

Both  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values from the exhaust gases varied widely amongst the three vehicles (Fig. 2). The  $\delta^{13}\text{C}$  value of  $\text{CH}_4$  emitted from CAR-66 ( $-21 \pm 1\text{‰}$ ) was similar to that of gasoline (range =  $-32$  to  $-26\text{‰}$ ; Aggarwal and Hinchee, 1991; Jackson et al., 1996; Silverman and Epstein, 1958); however,  $\delta^{13}\text{C}$   $\text{CH}_4$  values for the newer automobiles were  $^{13}\text{C}$ -enriched by more than  $15\text{‰}$  ( $-11 \pm 2\text{‰}$  for CAR-93 and  $-9 \pm 3\text{‰}$  for CAR-02). This trend agrees with a previous report of  $\delta^{13}\text{C}$   $\text{CH}_4$  values for automobile exhaust measured from 16 independent automobiles (Chanton et al., 2000; Fig. 3). A negative correlation between  $\delta^{13}\text{C}$  value and abundance also has been reported for CO in exhaust gases from automobiles (Tsunogai et al., 2003). Enrichment in  $^{13}\text{C}$  would be expected to occur as a result of incomplete combustion of  $\text{CH}_4$  within the engine or partial removal by the catalytic converter. A similar trend, but with larger variation was found in  $\delta\text{D}$   $\text{CH}_4$  values:  $-170 \pm 6\text{‰}$ ,  $-123 \pm 9\text{‰}$ , and  $-81 \pm 32\text{‰}$ , respectively, for CAR-66, CAR-93 and CAR-02 (Fig. 3). The  $\delta\text{D}$   $\text{CH}_4$  values from automobile exhaust are mostly within the range of  $\delta\text{D}$  values for petroleum hydrocarbons ( $\delta\text{D} = -175$  to  $-71\text{‰}$ ; Li et al., 2001; Pond et al., 2002) in contrast to biomass burning in which  $\text{CH}_4$  always exhibits a significant D depletion relative to fuel as a result of isotopic fractionation associated with incomplete combustion ( $\varepsilon_{\text{D}} = -130$  to  $-180\text{‰}$ ; Snover et al., 2000). However, the range of measured  $\delta\text{D}$   $\text{CH}_4$  values from exhaust gases is difficult to explain entirely

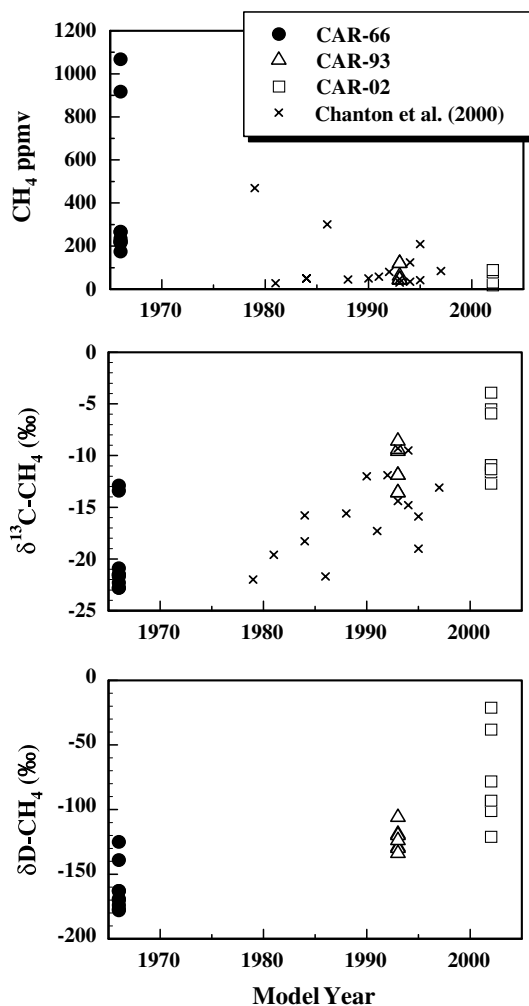


Fig. 3. Ranges of concentration, and stable carbon and hydrogen isotope compositions of  $\text{CH}_4$  in automobile exhaust in relation to model year of engine. Data from Chanton et al. (2000) are for  $\text{CH}_4$  in exhaust gas collected and analyzed from 16 automobiles between 1979 and 1997.

by  $\delta\text{D}$  variation in gasoline. It is also plausible to attribute some of the variation in  $\delta\text{D}$   $\text{CH}_4$  values to fractionation associated with removal of  $\text{CH}_4$  from exhaust by catalytic converters. This suggestion is supported by the observation that  $\delta\text{D}$   $\text{CH}_4$  values for CAR-66 are more negative and exhibit less variation than the new automobiles, especially CAR-02.

A strong inverse relationship between the log of  $\text{CH}_4$  content and  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of  $\text{CH}_4$  in automobile exhaust (Figs. 4 and 5) suggests that  $^{13}\text{C}$ - and D-enrichment in  $\text{CH}_4$  results from isotopic fractionation effects associated with a specific oxidation process, most probably catalytic oxidation over a metal surface. The magnitude of carbon and hydrogen kinetic isotopic fractionations for  $\text{CH}_4$  removal described by the rela-

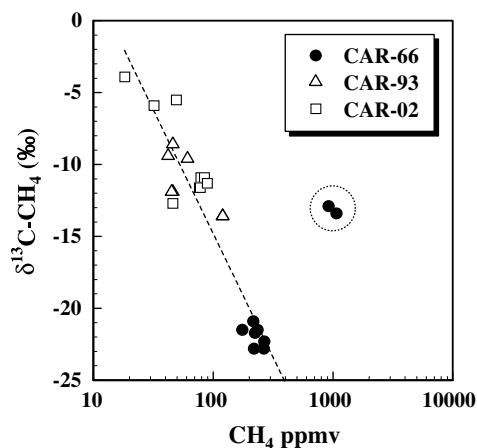


Fig. 4. Stable carbon isotope values versus concentration of CH<sub>4</sub> in automobile exhaust.

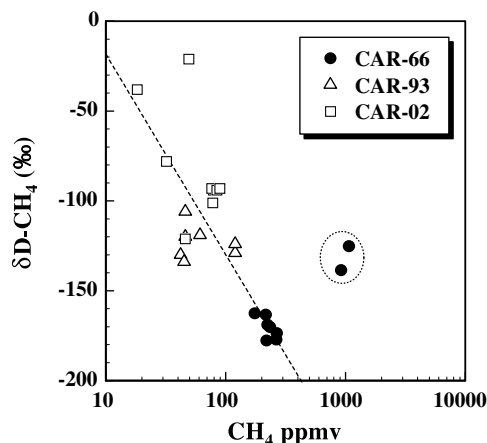


Fig. 5. Stable hydrogen isotope value versus concentration of CH<sub>4</sub> in automobile exhaust.

tionships between CH<sub>4</sub> abundance and δ<sup>13</sup>C and δD CH<sub>4</sub> values shown in Figs. 4 and 5, respectively, were calculated using the equations:

$$\varepsilon_{13C} = \left( \frac{k_{12}}{k_{13}} - 1 \right) \times 1000, \quad (2)$$

$$\varepsilon_D = \left( \frac{k_1}{k_2} - 1 \right) \times 1000, \quad (3)$$

where  $k$  is the rate constant for the different isotopomers. The fractionation factors were calculated using a Rayleigh distillation approach, which is defined by the equation:

$$\delta_t = \delta_i - \varepsilon \times \ln \left( \frac{M_t}{M_i} \right), \quad (4)$$

where  $M$  is CH<sub>4</sub> concentration and  $\delta$  is the stable isotope composition of CH<sub>4</sub>. The subscript  $i$  denotes the initial concentration and isotopic composition of CH<sub>4</sub>, and  $t$  represents the remaining data, with the exception of the two anomalous measurements for CAR-66 which were excluded from the calculation.

By using the average concentration and isotopic composition of CH<sub>4</sub> from CAR-66 (229 ppmv, δ<sup>13</sup>C = −21‰ and δD = −170‰) as  $M_i$  and  $\delta_i$  values (the isotope fractionation effect associated with partial catalytic removal of CH<sub>4</sub> can be neglected for CAR-66 because it lacks a catalytic converter),  $\varepsilon_C$  and  $\varepsilon_D$  are calculated to be  $+7.5 \pm 0.8‰$  and  $+48 \pm 9‰$ , respectively. The calculated isotopic fractionation factors agree well with other  $\varepsilon_C$  and  $\varepsilon_D$  values reported for processes involving partial oxidation of CH<sub>4</sub>, which typically result in <sup>13</sup>C and D enrichment and larger values of  $\varepsilon_D$  versus  $\varepsilon_C$  e.g., reaction with metal oxide (Kiyosu and Imaizumi, 1996), OH radicals (Cantrell et al., 1990; Saueressig et al., 2001), O(<sup>1</sup>D) (Saueressig et al., 2001), and bacterial methanotrophy (Bergamaschi et al., 1998; Coleman et al., 1981)).

The average δ<sup>13</sup>C and δD values of CH<sub>4</sub> in automobile exhaust gases during the period 1993–1994 were estimated from average δ<sup>13</sup>C and δD CH<sub>4</sub> values of exhaust from CAR-66 (188 ppmv, δ<sup>13</sup>C = −19.3‰, δD = −170‰), CAR-93 (78 ppmv, δ<sup>13</sup>C = −12.0‰, δD = −123‰) and stable isotope data reported by Chanton et al. (2000). Assuming that 20% of automobiles driven in Nagoya lack a catalytic converter, the average δ<sup>13</sup>C and δD CH<sub>4</sub> values of automobile exhaust are estimated to be δ<sup>13</sup>C = −13.5 ± 1.5‰ and δD = −133 ± 10‰. These δ values are considerably more positive than those for the other three main sources of CH<sub>4</sub> in urban areas of Nagoya which are described below.

### 3.1.2. Natural gas leakage

A natural gas sample collected from the gas mains at Nagoya University was analyzed to determine the stable isotope composition of CH<sub>4</sub> in leakage from gas pipelines in Nagoya City. The measured δ<sup>13</sup>C and δD CH<sub>4</sub> values were −42.5‰ and −168‰, respectively, which are in good agreement with values typically reported for thermogenic CH<sub>4</sub> from natural gas wells (Schoell, 1980, 1988).

### 3.1.3. Rice paddy emissions

The δ<sup>13</sup>C and δD values of microbial CH<sub>4</sub> from rice cultivation in Nagoya were estimated using data reported from previous studies of Japanese rice paddies (Nakagawa et al., 2002; Uzaki et al., 1991). The δ<sup>13</sup>C value of CH<sub>4</sub> emitted from Nagoya rice paddies was estimated to be −63 ± 4‰ after considering the influence of isotope fractionation associated with secondary processes (i.e., CH<sub>4</sub> oxidation and transport; Bergamaschi, 1997; Chanton et al., 1997; Tyler et al., 1997) on the



primary  $\delta^{13}\text{C}$  signatures of  $\text{CH}_4$  produced within the soil (Uzaki et al., 1991).

The  $\delta\text{D}$  value of  $\text{CH}_4$  from rice paddies, known to be related to that of co-existing water (Sugimoto and Wada, 1995; Waldron et al., 1999), was calculated using the relationship reported by Nakagawa et al. (2002):

$$\delta\text{D}_{\text{CH}_4} = 0.54 \times \delta\text{D}_{\text{H}_2\text{O}} - 314\text{‰} \quad (5)$$

by substituting the  $\delta\text{D}$  value of floodwater ( $-35$  to  $-50\text{‰}$ ; Nakagawa et al., 2002) into the equation and by considering isotope effects associated with secondary processes (Bergamaschi, 1997; Tyler et al., 1997). The  $\delta\text{D}$  value of  $\text{CH}_4$  emitted from rice paddies in Nagoya was estimated to be  $-320 \pm 15\text{‰}$ .

### 3.1.4. Landfill emissions

We used the  $\delta^{13}\text{C}$  values of  $\text{CH}_4$  emissions from landfills, which have been reported by Bergamaschi et al. (1998) and Chanton et al. (1999) to range from  $-58$  to

$-49\text{‰}$ ;  $\delta\text{D}$   $\text{CH}_4$  values typically lie in the range  $-312$  to  $-290\text{‰}$  (Bergamaschi et al., 1998; Chanton and Liptay, 2000; Levin et al., 1993; Wahlen, 1993). These  $\delta$  values indicate slight enrichment in  $^{13}\text{C}$  and D relative to the other microbial sources of  $\text{CH}_4$ .

### 3.2. Comparison with atmospheric $\text{CH}_4$

Stable carbon and hydrogen isotope compositions and mixing ratios for  $\text{CH}_4$  in air samples collected at the three sites (Stn-H, Stn-N, and Stn-T) are listed in Table 2. At all sites, the concentration of  $\text{CH}_4$  was higher than background values reported at a similar latitude (summer, 1.73 ppmv and winter, 1.78 ppmv; Dlugokencky et al., 1994), indicating that  $\text{CH}_4$  flux was occurring from local sources. The maximum contribution of  $\text{CH}_4$  from local sources was estimated to be 26% at Stn-H in the winter, 25% and 33% at Stn-N in summer and winter, respectively, and 94% at Stn-T in the

Table 2  
Mixing ratio and stable isotope composition of atmospheric  $\text{CH}_4$  in Nagoya

| Date                             | Time  | $\text{CH}_4$ mixing ratio <sup>b</sup> (ppmv) | $\delta^{13}\text{C}$ $\text{CH}_4$ <sup>b</sup> (‰) | $\delta\text{D}$ $\text{CH}_4$ <sup>a</sup> (‰) |
|----------------------------------|-------|------------------------------------------------|------------------------------------------------------|-------------------------------------------------|
| <i>Stn-H (Higashi-Sakura)</i>    |       |                                                |                                                      |                                                 |
| 93/11/04                         | 5:42  | $1.915 \pm 0.005$                              | -46.9                                                | -106                                            |
| 93/11/10                         | 5:15  | $1.910 \pm 0.005$                              | -46.1                                                | -104                                            |
| 93/11/10                         | 14:20 | $2.170 \pm 0.007$                              | -44.9                                                |                                                 |
| 93/11/16                         | 5:15  | $2.024 \pm 0.005$                              | -45.8                                                | -113                                            |
| 93/11/17                         | 11:44 | $2.103 \pm 0.007$                              | -45.3                                                | -112                                            |
| 93/11/22                         | 14:03 | $1.936 \pm 0.009$                              | -46.0                                                | -103                                            |
| 93/11/25                         | 5:30  | $2.234 \pm 0.008$                              | -45.8                                                | -120                                            |
| 93/12/15                         | 11:07 | $1.891 \pm 0.006$                              | -45.5                                                | -101                                            |
| 93/12/16                         | 6:12  | $1.944 \pm 0.005$                              | -45.7                                                |                                                 |
| <i>Stn-T (Tomida)</i>            |       |                                                |                                                      |                                                 |
| 94/07/15                         | 4:07  | $3.350 \pm 0.061$                              | -54.5                                                | -188                                            |
| 94/07/18                         | 4:33  | $2.324 \pm 0.030$                              | -49.9                                                | -139                                            |
| 94/07/20                         | 4:11  | $2.449 \pm 0.022$                              | -47.2                                                | -131                                            |
| 94/07/25                         | 4:22  | $1.856 \pm 0.018$                              | -45.9                                                | -87                                             |
| 94/07/28                         | 4:34  | $2.220 \pm 0.030$                              | -47.1                                                | -128                                            |
| 94/08/01                         | 14:19 | $1.933 \pm 0.015$                              | -46.0                                                | -90                                             |
| 94/08/08                         | 4:05  | $2.203 \pm 0.021$                              | -48.4                                                | -124                                            |
| 94/08/08                         | 14:45 | $1.850 \pm 0.013$                              | -46.0                                                | -90                                             |
| <i>Stn-N (Nagoya-University)</i> |       |                                                |                                                      |                                                 |
| Summer                           |       |                                                |                                                      |                                                 |
| 93/07/03                         | 14:04 | $1.747 \pm 0.011$                              | -47.2                                                | -99                                             |
| 93/07/04                         | 4:10  | $1.876 \pm 0.016$                              | -46.4                                                | -108                                            |
| 93/07/06                         | 4:00  | $2.153 \pm 0.018$                              | -47.9                                                | -130                                            |
| 93/07/16                         | 14:11 | $1.757 \pm 0.009$                              | -46.1                                                | -97                                             |
| Winter                           |       |                                                |                                                      |                                                 |
| 93/11/06                         | 5:26  | $2.362 \pm 0.008$                              | -45.7                                                | -125                                            |
| 93/11/06                         | 5:50  | $2.206 \pm 0.006$                              | -46.0                                                | -125                                            |
| 94/01/08                         | 6:29  | $1.924 \pm 0.006$                              | -46.5                                                | -103                                            |
| 94/02/08                         | 6:09  | $2.027 \pm 0.005$                              | -46.2                                                | -103                                            |
| 94/02/11                         | 12:40 | $1.849 \pm 0.005$                              | -46.3                                                | -95                                             |

<sup>a</sup> This work.

<sup>b</sup> Moriizumi et al. (1995, 1998).

summer. At Stn-N and Stn-T, the concentrations of CH<sub>4</sub> generally were higher at night than during the day because of the thin nocturnal inversion layer which traps CH<sub>4</sub> emitted from local sources. During the daytime, active mixing with the free troposphere dilutes enhanced levels of CH<sub>4</sub> from local emissions to that of the free troposphere. In the case of Stn-H, which is located in downtown Nagoya City, CH<sub>4</sub> concentration during the daytime was similar to the level observed at night, suggesting diurnal variation in source strength and a relatively higher contribution of anthropogenic emissions in the daytime.

The  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of atmospheric CH<sub>4</sub> varied to some extent at each site (Table 2). The largest variations of 8.6‰ for  $\delta^{13}\text{C}$  and 101‰ for  $\delta\text{D}$  occurred at Stn-T. Fig. 6 shows the linear relationships that existed between the isotope composition of atmospheric methane and its reciprocal concentration at the study sites. At all sites,  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of atmospheric CH<sub>4</sub> approached the known tropospheric background values ( $\delta^{13}\text{C} = -47.2$  to  $-47.6$ ‰ and  $\delta\text{D} = -80$  to  $-95$ ‰; Quay et al., 1999) when the CH<sub>4</sub> mixing ratio decreased to background levels. The  $\delta^{13}\text{C}$  and  $\delta\text{D}$  CH<sub>4</sub> values varied at higher mixing ratios because CH<sub>4</sub> emissions from local sources have different stable isotope compositions from background tropospheric CH<sub>4</sub>.

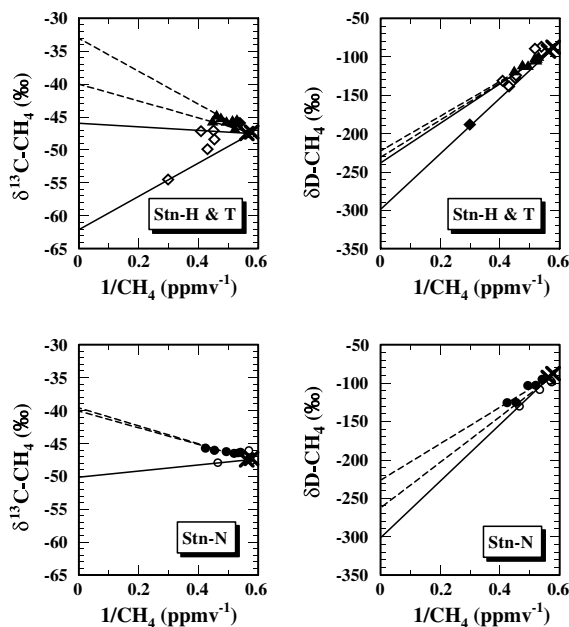


Fig. 6. Relationships between inverse of mixing ratio of atmospheric CH<sub>4</sub> versus  $\delta^{13}\text{C}$  and  $\delta\text{D}$  CH<sub>4</sub> values at Stn-H (winter: solid triangle), Stn-T (summer: open rhombus), and Stn-N (summer: open circle, winter: solid circle). Solid and dotted lines represent range of mixing lines between isotopic composition of local CH<sub>4</sub> sources and tropospheric background (cross; Quay et al., 1999) in summer and winter, respectively.

Atmospheric CH<sub>4</sub> was D-depleted at higher concentrations, suggesting that accumulated CH<sub>4</sub> from local sources had low  $\delta\text{D}$  values. This finding is consistent with the D-depleted values for CH<sub>4</sub> discussed in Section 3.1. A negative correlation between  $\delta\text{D}$  CH<sub>4</sub> values and CH<sub>4</sub> abundance also can be explained by isotopic fractionation associated with CH<sub>4</sub> oxidation by OH radicals in the troposphere (Cantrell et al., 1990). However, this process is unlikely to be important in short term observations because of the  $\sim 10$  year lifetime of CH<sub>4</sub> in the troposphere. Therefore, simple mixing between local sources with low  $\delta\text{D}$  values and background CH<sub>4</sub> is the most plausible explanation of variations in atmospheric  $\delta\text{D}$  CH<sub>4</sub> values in Nagoya.

The relationship between the stable carbon isotope composition of CH<sub>4</sub> and its concentration differed among sites and seasons, suggesting that the predominant sources of atmospheric CH<sub>4</sub> differed spatially and temporally. Generally,  $^{13}\text{C}$  enrichment with increasing CH<sub>4</sub> concentration was observed in winter, whereas  $^{13}\text{C}$  depletion was observed in summer. This finding suggests that fossil CH<sub>4</sub> sources, such as natural gas leakage or automobile exhaust, were the dominant sources of atmospheric CH<sub>4</sub> in winter, while contributions from biogenic sources increased during the summer. A quantitative evaluation of this speculation is provided below.

If simple mixing processes determine the CH<sub>4</sub> distribution in the atmosphere, then a linear relationship should exist between the isotopic compositions of CH<sub>4</sub> and the reciprocal of CH<sub>4</sub> mixing ratios according the equation:

$$\delta = \delta_s + \frac{C_B \times (\delta_B - \delta_s)}{C} \quad (6)$$

where  $C$  and  $\delta$  denote the concentration and isotopic composition of CH<sub>4</sub>, respectively, the subscripts B and S denote background and local source values for CH<sub>4</sub>, respectively. Thus,  $\delta_s$  is the average stable isotope composition of local sources calculated by weighting the stable isotope signature of individual sources ( $\delta_i$ ) by the contribution ratio ( $f_i$ ):

$$\delta_s = \sum_{i=1}^n f_i \times \delta_i. \quad (7)$$

The mean isotopic composition of CH<sub>4</sub> from local sources ( $\delta_s$ ) can be calculated from Eq. (6) and then Eq. (7) can be used to estimate the contribution of individual sources.

The mean  $\delta^{13}\text{C}$  and  $\delta\text{D}$  CH<sub>4</sub> values of local sources were calculated for individual atmospheric samples using Eq. (6); however, application of the equation was limited to air samples having a CH<sub>4</sub> contribution of at least 15% from local anthropogenic sources (Table 3). The concentration and isotope compositions of background CH<sub>4</sub> used in the calculation were 1.73 ppmv,



Table 3

Calculated mean  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of local  $\text{CH}_4$  emissions and estimated contributions of three major anthropogenic sources to atmospheric  $\text{CH}_4$  at Stn-H, Stn-T and Stn-N in Nagoya

| Site                  | $\delta^{13}\text{C}$ and $\delta\text{D}$ of $\text{CH}_4$ |                      | Estimated contributions (%) |                      |                                     |
|-----------------------|-------------------------------------------------------------|----------------------|-----------------------------|----------------------|-------------------------------------|
|                       | $\delta^{13}\text{C}$ (‰)                                   | $\delta\text{D}$ (‰) | Automobile exhaust          | Natural gas pipeline | Microbial (landfill and rice paddy) |
| <i>Stn-H (winter)</i> |                                                             |                      |                             |                      |                                     |
| 93/11/17 (11:44)      | $-32.9 \pm 0.8$                                             | $-220 \pm 16$        | 51 (42–53)                  | 0 (0–24)             | 49 (33–53)                          |
| 93/11/25 (5:30)       | $-39.2 \pm 0.6$                                             | $-228 \pm 12$        | 31 (22–38)                  | 15 (0–42)            | 54 (37–64)                          |
| <i>Stn-T (summer)</i> |                                                             |                      |                             |                      |                                     |
| 94/07/15 (4:07)       | $-62.2 \pm 0.2$                                             | $-296 \pm 4$         | 0 (0–11)                    | 4 (0–24)             | 96 (77–100)                         |
| 94/07/18 (4:33)       | $-57.4 \pm 0.5$                                             | $-287 \pm 9$         | 5 (0–20)                    | 15 (0–27)            | 80 (68–88)                          |
| 94/07/20 (4:11)       | $-46.8 \pm 0.4$                                             | $-237 \pm 8$         | 21 (11–39)                  | 29 (0–41)            | 50 (41–64)                          |
| 94/07/28 (4:34)       | $-46.2 \pm 0.5$                                             | $-269 \pm 11$        | 26 (13–26)                  | 0 (0–15)             | 73 (64–83)                          |
| 94/08/08 (4:05)       | $-52.4 \pm 0.6$                                             | $-258 \pm 11$        | 10 (0–29)                   | 29 (0–43)            | 61 (49–78)                          |
| <i>Stn-N (winter)</i> |                                                             |                      |                             |                      |                                     |
| 93/11/06 (5:26)       | $-40.3 \pm 0.5$                                             | $-226 \pm 9$         | 26 (18–32)                  | 23 (10–46)           | 51 (36–58)                          |
| 93/11/06 (5:50)       | $-39.9 \pm 0.6$                                             | $-261 \pm 13$        | 28 (24–36)                  | 0 (0–17)             | 72 (58–72)                          |
| <i>Stn-N (summer)</i> |                                                             |                      |                             |                      |                                     |
| 93/07/06 (4:00)       | $-50.4 \pm 0.6$                                             | $-304 \pm 12$        | 11 (0–11)                   | 0 (0–7)              | 89 (84–100)                         |

$\delta^{13}\text{C} = -47.3 \pm 0.1\text{‰}$ ,  $\delta\text{D} = -87.5 \pm 2.5\text{‰}$  for summer and  $1.78 \text{ ppmv}$ ,  $\delta^{13}\text{C} = -47.5 \pm 0.1\text{‰}$ ,  $\delta\text{D} = -92.5 \pm 2.5\text{‰}$  for winter (Quay et al., 1999).

The stable isotope compositions of local sources ( $\delta_s$ ) differed amongst sites and seasons (Table 3), indicating that the predominant  $\text{CH}_4$  sources were likely to be different between sites and were temporally variable at least at Stn-N. In general, anthropogenic sources appeared to be more  $^{13}\text{C}$ - and D-enriched during winter months. The mean  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values of local sources ( $\delta_s$ ) at each site are shown in Fig. 7 plotted along

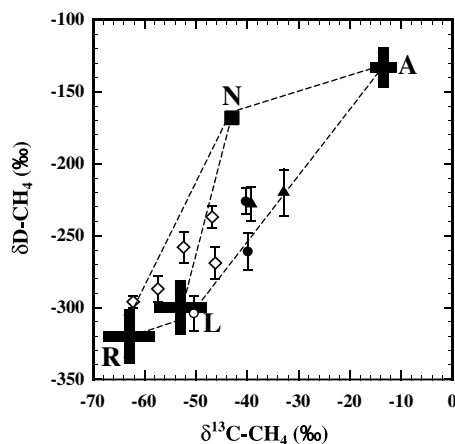


Fig. 7. Stable carbon and hydrogen isotope values for anthropogenic  $\text{CH}_4$  emissions at Stn-H (winter: solid triangle), Stn-T (summer: open rhombus), and Stn-N (summer: open circle). The solid squares denote the stable isotope compositions of major individual  $\text{CH}_4$  sources in Nagoya (automobile exhaust (A), leakage of natural gas (N), rice paddies (R), and landfills (L)).

with the stable isotope compositions of the four major local sources of  $\text{CH}_4$  estimated in Section 3.1. The estimated mean  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values of local sources in winter (solid circle and triangle in Fig. 7) do not fall within the field delineated by the three vertices of  $\text{CH}_4$  from natural gas leakage, rice paddy emissions and landfill flux, which suggests a significant contribution from a  $^{13}\text{C}$ -enriched source. Generally, exhaust gas from automobiles is neglected as a source of atmospheric  $\text{CH}_4$  in the global budget; however, these results indicate that this source of  $^{13}\text{C}$ -enriched  $\text{CH}_4$  needs to be considered for urban atmospheric budgets, especially in large cities where traffic is heavy.

The contribution from individual local sources was calculated by substituting the estimated mean  $\delta$  values of local sources at each site (determined in Section 3.1) into Eq. (7). The number of unknown parameters (i.e., the number of local sources) was reduced from four to three [because only two variables ( $\delta^{13}\text{C}$  and  $\delta\text{D}$ ) were available] by averaging  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values for rice paddy and landfill emissions to form a “microbial source”. It was difficult to distinguish the relative contribution of rice paddy and landfill flux to the “microbial source” during summer months, leading to larger errors in its  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values ( $\delta^{13}\text{C}$ :  $-53 \pm 5\text{‰}$  to  $-63 \pm 5\text{‰}$ ,  $\delta\text{D}$ :  $-300 \pm 15\text{‰}$  to  $-320 \pm 15\text{‰}$ ); however, the contribution from rice agriculture could be neglected during the winter because Japanese rice paddies typically are not flooded at that time of year and hence, microbial  $\text{CH}_4$  should not be produced. The estimated contributions from the three sources at each site are shown in Table 3. At Stn-H, approximately 50% of anthropogenic  $\text{CH}_4$  emissions was derived from fossil carbon sources (i.e., natural gas leakage and exhaust gases from automobiles), while at Stn-T the majority of  $\text{CH}_4$  was

released from microbial sources (most probably from rice paddies). The estimated contribution of fossil ( $^{14}\text{C}$ -free)  $\text{CH}_4$  to the urban and suburban atmosphere in Nagoya reported in previous studies supports these conclusions (Moriizumi et al., 1998). At Stn-N, the contribution of microbial  $\text{CH}_4$  from landfills located north-east of central Nagoya (Ito et al., 2001) was larger than fossil carbon sources in summer. In winter, the relative contribution of fossil carbon was slightly higher, but the “microbial source” still dominated anthropogenic emissions. At all sites, a significant contribution of  $\text{CH}_4$  from automobile exhaust gas was present.

### 3.3. Implications for isotope mass balance calculations for $\text{CH}_4$ in urban areas

Exclusion of automobile exhaust emissions in isotope mass balance evaluations of atmospheric  $\text{CH}_4$  sources in urban areas results in overestimation of contributions from natural gas leakage and underestimation of the flux from microbial sources. Moreover, the use of only  $^{13}\text{C}$  as an isotopic tracer can greatly limit the accuracy of estimating the relative contribution of emissions from different sources. For example, if the mass balance calculation based upon Eq. (7) is performed with only  $\delta^{13}\text{C}$  values for natural gas leakage and landfill emissions in winter, then the results suggests that 100% of anthropogenic  $\text{CH}_4$  emissions at Stn-N are derived from natural gas leakage and negligible release occurs from landfills. Such a result contradicts previous studies which have demonstrated that landfills are a major source of atmospheric  $\text{CH}_4$  in this urban area (Ito et al., 2001). However, including automobile exhaust in the two dimensional mass balance calculation and using both  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values yields the more accurate finding that landfills are an important local source of atmospheric  $\text{CH}_4$ .

## 4. Conclusion

The stable isotope compositions of  $\text{CH}_4$  from automobile exhaust were determined to evaluate the contribution of this anthropogenic source to the atmosphere in urban areas of Nagoya, Japan, using an isotope mass balance approach that involved both  $^{13}\text{C}$  and  $\text{D}$  as isotopic tracers. Large variations were observed in both  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of  $\text{CH}_4$  in automobile exhaust ( $\delta^{13}\text{C} = -22.8$  to  $-3.9\text{‰}$  and  $\delta\text{D} = -178$  to  $-21\text{‰}$ ) with mean  $\delta$  values showing strong  $^{13}\text{C}$  and  $\text{D}$  enrichment compared to stable isotope signatures from other local anthropogenic sources. Methane in exhaust gas from recent automobiles was more  $^{13}\text{C}$  and  $\text{D}$  enriched compared to emissions from older automobiles, presumably because of isotope fractionation effects associated with

oxidation over metal surfaces in catalytic converters. This finding suggests that average  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values for automobile exhaust might be shifting to more positive values with time as a result of improved efficiency of catalytic converters in new automobiles, possibly leading to greater spatial and temporal variations globally in the stable isotope signature of this poorly constrained  $\text{CH}_4$  emission source.

Based upon  $\delta^{13}\text{C}$  and  $\delta\text{D}$  values of  $\text{CH}_4$  in automobile exhaust determined in this study and estimates of average  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values for different local sources (leakage from natural gas pipelines, and emissions from rice paddies and landfills), the relative contributions of anthropogenic  $\text{CH}_4$  sources to the atmosphere in urban areas of Nagoya were determined using mass balance calculations. Results indicate that landfills and automobile exhaust contributed significant quantities to the atmosphere in the study area. Use of both  $\delta^{13}\text{C}$  and  $\delta\text{D}$   $\text{CH}_4$  values enabled a more accurate assessment of the relative contributions from different anthropogenic sources to the atmosphere in this urban environment in comparison to calculations based solely upon  $\delta^{13}\text{C}$   $\text{CH}_4$  values.

## Acknowledgement

We thank Drs. D. Lowry (University of London) and E.R.C. Hornibrook (University of Bristol) for their constructive comments. We also thank Dr. J.P. Chanton (Florida State University) for providing the stable carbon isotope data for  $\text{CH}_4$  from automobile exhaust for our calculations. We are grateful to Drs. Y. Hachisu, J. Yamaguchi, and A. Ijiri (Hokkaido University) for their cooperation in the automobile exhaust experiments. This work was partly supported by the following grants: Nissan Science Foundation Society for the Promotion of Science, MEXT Grant-in Aid for Scientific Research Nos. 13740313 and 14048202, MEXT Grant-in Aid for the 21st Century COE Program “Neo-Science of Natural History” at Hokkaido University, MEXT Special Coordination Fund for the “Archaean Park” Project, and NEDO Proposal Based Research and Development Project (ID 02A53002d).

## References

- Aggarwal, P.K., Hinchey, R.E., 1991. Monitoring insitu biodegradation of hydrocarbons by using stable carbon isotopes. *Environmental Science and Technology* 25, 1178–1180.
- Bergamaschi, P., 1997. Seasonal variations of stable hydrogen and carbon isotope ratios in methane from a Chinese rice paddy. *Journal of Geophysical Research* 102, 25383–25393.

- Bergamaschi, P., Lubina, C., Konigstedt, R., Fischer, H., Veltkamp, A.C., Zwaagstra, O., 1998. Stable isotopic signatures ( $\delta^{13}\text{C}$ ,  $\delta\text{D}$ ) of methane from European landfill sites. *Journal of Geophysical Research* 103, 8251–8265.
- Cantrell, C.A., Shetter, R.E., McDaniel, A.H., Calvert, J.G., Davidson, J.A., Lowe, D.C., Tyler, S.C., Cicerone, R.J., Greenberg, J.P., 1990. Carbon kinetic isotope effect in the oxidation of methane by the hydroxyl radical. *Journal of Geophysical Research* 95, 22455–22462.
- Chanton, J., Liptay, K., 2000. Seasonal variation in methane oxidation in a landfill cover soil as determined by an in situ stable isotope technique. *Global Biogeochemical Cycles* 14, 51–60.
- Chanton, J.P., Rutkowski, C.M., Mosher, B., 1999. Quantifying methane oxidation from landfills using stable isotope analysis of downwind plumes. *Environmental Science & Technology* 33, 3755–3760.
- Chanton, J.P., Rutkowski, C.M., Schwartz, C.C., Ward, D.E., Boring, L., 2000. Factors influencing the stable carbon isotopic signature of methane from combustion and biomass burning. *Journal of Geophysical Research* 105, 1867–1877.
- Chanton, J.P., Whiting, G.J., Blair, N.E., Lindau, C.W., Bollich, P.K., 1997. Methane emission from rice: Stable isotopes, diurnal variations, and  $\text{CO}_2$  exchange. *Global Biogeochemical Cycles* 11, 15–27.
- Cicerone, R.J., Oremland, R.S., 1988. Biogeochemical aspects of atmospheric methane. *Global Biogeochemical Cycles* 2, 299–327.
- Coleman, D.D., Risatti, J.B., Schoell, M., 1981. Fractionation of carbon and hydrogen isotopes by methane-oxidizing bacteria. *Geochimica et Cosmochimica Acta* 45, 1033–1037.
- Coleman, M.L., Shepherd, T.J., Durham, J.J., Rouse, J.E., Moore, G.R., 1982. Reduction of water with zinc for hydrogen isotope analysis. *Analytical Chemistry* 54, 993–995.
- Craig, H., Chou, C.C., 1982. Methane – the record in polar ice cores. *Geophysical Research Letters* 9, 1221–1224.
- Dlugokencky, E.J., Steele, L.P., Lang, P.M., Masarie, K.A., 1994. The growth-rate and distribution of atmospheric methane. *Journal of Geophysical Research* 99, 17021–17043.
- Etheridge, D.M., Steele, L.P., Francey, R.J., Langenfelds, R.L., 1998. Atmospheric methane between 1000 AD and present: evidence of anthropogenic emissions and climatic variability. *Journal of Geophysical Research* 103, 15979–15993.
- IPCC, 1995. In: Houghton, J.T. (Ed.), *Climate Change 1994: Radiative Forcing of Climate Change and an Evolution of the IPCC IS93 Emission Scenarios*. Cambridge University Press, Cambridge.
- Ito, A., Takahashi, I., Nagata, Y., Chiba, K., Haraguchi, H., 2000. The long-term evolutions and the regional characteristics of atmospheric methane concentrations in Nagoya, 1983–1997. *Science of the Total Environment* 263, 37–45.
- Ito, A., Takahashi, I., Nagata, Y., Chiba, K., Haraguchi, H., 2001. Spatial and temporal characteristics of urban atmospheric methane in Nagoya City, Japan: an assessment of the contribution from regional landfills. *Atmospheric Environment* 35, 3137–3144.
- Jackson, A.W., Pardue, J.H., Araujo, R., 1996. Monitoring crude oil mineralization in salt marshes: use of stable carbon isotope ratios. *Environmental Science and Technology* 30, 1139–1144.
- Kiyosu, Y., Imaizumi, S., 1996. Carbon and hydrogen isotope fractionation during oxidation of methane by metal oxides at temperatures from 400 °C to 530 °C. *Chemical Geology* 133, 279–287.
- Kuc, T., Rozanski, K., Zimnoch, M., Necki, J.M., Korus, A., 2003. Anthropogenic emissions of  $\text{CO}_2$  and  $\text{CH}_4$  in an urban environment. *Applied Energy* 75, 193–203.
- Lacroix, A.V., 1993. Unaccounted-for sources of fossil and isotopically-enriched methane and their contribution to the emissions inventory – a review and synthesis. *Chemosphere* 26, 507–557.
- Levin, I., Bergamaschi, P., Dorr, H., Trapp, D., 1993. Stable isotopic signature of methane from major sources in Germany. *Chemosphere* 26, 161–177.
- Levin, I., Glatzel-Mattheier, H., Marik, T., Cuntz, M., Schmidt, M., Worthy, D.E., 1999. Verification of German methane emission inventories and their recent changes based on atmospheric observations. *Journal of Geophysical Research-Atmospheres* 104, 3447–3456.
- Li, M.W., Huang, Y.S., Obermajer, M., Jiang, C.Q., Snowden, L.R., Fowler, M.G., 2001. Hydrogen isotopic compositions of individual alkanes as a new approach to petroleum correlation: case studies from the Western Canada Sedimentary Basin. *Organic Geochemistry* 32, 1387–1399.
- Lowry, D., Holmes, C.W., Rata, N.D., O'Brien, P., Nisbet, E.G., 2001. London methane emissions: use of diurnal changes in concentration and  $\delta^{13}\text{C}$  to identify urban sources and verify inventories. *Journal of Geophysical Research* 106, 7427–7448.
- Moriizumi, J., Nagamine, K., Iida, T., Ikebe, Y., 1996. Estimation of areal flux of atmospheric methane in an urban area of Nagoya, Japan, inferred from atmospheric radon-222 data. *Atmospheric Environment* 30, 1543–1549.
- Moriizumi, J., Nagamine, K., Iida, T., Ikebe, Y., 1998. Carbon isotopic analysis of atmospheric methane in urban and suburban areas: fossil and non-fossil methane from local sources. *Atmospheric Environment* 32, 2947–2955.
- Moriizumi, J., Nagamine, K., Iida, T., Ikebe, Y., Nakamura, T., 1995. Preparation of trace amount of atmospheric methane for measuring  $^{14}\text{C}$  isotopic ratio. *Chikyukagaku Geochemistry* 29, 99–111.
- Nakagawa, F., Yoshida, N., Sugimoto, A., Wada, E., Yoshioka, T., Ueda, S., Vijarnsorn, P., 2002. Stable isotope and radiocarbon compositions of methane emitted from tropical rice paddies and swamps in Southern Thailand. *Biogeochemistry* 61, 1–19.
- Pearman, G.I., Etheridge, D., Desilva, F., Fraser, P.J., 1986. Evidence of changing concentrations of atmospheric  $\text{CO}_2$ ,  $\text{N}_2\text{O}$  and  $\text{CH}_4$  from air bubbles in antarctic ice. *Nature* 320, 248–250.
- Pond, K.L., Huang, Y.S., Wang, Y., Kulpa, C.F., 2002. Hydrogen isotopic composition of individual *n*-alkanes as an intrinsic tracer for bioremediation and source identification of petroleum contamination. *Environmental Science and Technology* 36, 724–728.
- Quay, P., Stutsman, J., Wilbur, D., Snover, A., Dlugokencky, E., Brown, T., 1999. The isotopic composition of atmospheric methane. *Global Biogeochemical Cycles* 13, 445–461.

- Rasmussen, R.A., Khalil, M.A.K., 1984. Atmospheric methane in the recent and ancient atmospheres – concentrations, trends, and interhemispheric gradient. *Journal of Geophysical Research* 89, 1599–1605.
- Saueressig, G., Crowley, J.N., Bergamaschi, P., Bruhl, C., Brenninkmeijer, C.A.M., Fischer, H., 2001. Carbon 13 and D kinetic isotope effects in the reactions of CH<sub>4</sub> with O(<sup>1</sup>D) and OH: new laboratory measurements and their implications for the isotopic composition of stratospheric methane. *Journal of Geophysical Research* 106, 23127–23138.
- Schoell, M., 1980. The hydrogen and carbon isotopic composition of methane from natural gases of various origins. *Geochimica et Cosmochimica Acta* 44, 649–661.
- Schoell, M., 1988. Multiple origins of methane in the Earth. *Chemical Geology* 71, 1–10.
- Silverman, S.R., Epstein, S., 1958. Carbon isotopic compositions of petroleum and other sedimentary organic materials. *Bulletin of the American Association of Petroleum Geologists* 42, 998.
- Snover, A.K., Quay, P.D., Hao, W.M., 2000. The D/H content of methane emitted from biomass burning. *Global Biogeochemical Cycles* 14, 11–24.
- Stauffer, B., Fischer, G., Neftel, A., Oeschger, H., 1985. Increase of atmospheric methane recorded in antarctic ice core. *Science* 229, 1386–1388.
- Sugimoto, A., Wada, E., 1995. Hydrogen isotopic composition of bacterial methane – CO<sub>2</sub>/H<sub>2</sub> reduction and acetate fermentation. *Geochimica et Cosmochimica Acta* 59, 1329–1337.
- Tsunogai, U., Hachisu, Y., Komatsu, D.D., Nakagawa, F., Gamo, T., Akiyama, K., 2003. An updated estimation of the stable carbon and oxygen isotopic compositions of automobile CO emissions. *Atmospheric Environment* 37, 4901–4910.
- Tsunogai, U., Nakagawa, F., Komatsu, D.D., Gamo, T., 2002. Stable carbon and oxygen isotopic analysis of atmospheric carbon monoxide using continuous-flow isotope ratio MS by isotope ratio monitoring of CO. *Analytical Chemistry* 74, 5695–5700.
- Tsunogai, U., Yoshida, N., Ishibashi, J., Gamo, T., 2000. Carbon isotopic distribution of methane in deep-sea hydrothermal plume, Myojin Knoll Caldera, Izu-Bonin arc: implications for microbial methane oxidation in the oceans and applications to heat flux estimation. *Geochimica et Cosmochimica Acta* 64, 2439–2452.
- Tyler, S.C., Bilek, R.S., Sass, R.L., Fisher, F.M., 1997. Methane oxidation and pathways of production in a Texas paddy field deduced from measurements of flux,  $\delta^{13}\text{C}$ , and  $\delta\text{D}$  of CH<sub>4</sub>. *Global Biogeochemical Cycles* 11, 323–348.
- Uzaki, M., Mizutani, H., Wada, E., 1991. Carbon isotope composition of CH<sub>4</sub> from rice paddies in Japan. *Biogeochemistry* 13, 159–175.
- Wahlen, M., 1993. ‘bal methane cycle. *Annual Review of Earth and Planetary Sciences* 21, 407–426.
- Waldron, S., Lansdown, J.M., Scott, E.M., Fallick, A.E., Hall, A.J., 1999. The global influence of the hydrogen isotope composition of water on that of bacteriogenic methane from shallow freshwater environments. *Geochimica et Cosmochimica Acta* 63, 2237–2245.
- Yamada, K., Ozaki, Y., Nakagawa, F., Tanaka, M., Yoshida, N., 2003. An improved method for measurement of the hydrogen isotope ratio of atmospheric methane and its application to a Japanese urban atmosphere. *Atmospheric Environment* 37, 1975–1982.